A Provocation: Visualizing Modeled Social Networks

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Abstract: Network scholarship regularly leverages visualizations. They are thought to enhance readers' recall, interpretation, and simplification of scholar's intended takeaways. However, by representing the specifics of single empirical cases, these visualization strategies may not optimize how well they accomplish our scientific purposes. In particular, they may: (1) be overly complex in ways that obfuscate what we can learn from them, (2) draw attention to the particularities of the case, rather than the general principles the case is examined to characterize, (3) be difficult to precisely replicate, raising questions about consistency in interpretation, and (4) introduce potential risks that could violate research ethical standards. To address these, this paper provokes the field to consider whether presenting *modeled* network results–in lieu of empirical cases–retains the aims of network visualization, while potentially overcoming some of the limitations of existing practice.

NOTE: This is very much a fledgling idea that isn't yet ready for citation/sharing. If you're interested in helping me bring this to life, please reach out. Otherwise, for now, think of it as a musing... Also, I'd initially envisioned submitting as a Viz paper to *Socius*, and have since submitted it to *Social Networks* for their theory special issue (which has a submission timeline of Nov 1.

Social network research relies heavily on data visualization (Freeman 2004). These visualizations can serve to characterize, simplify, and focus theoretical or empirical claims (Moody et al. 2005; Rawlings et al. 2023). Visualizations are generally more readily digestible than tables of seemingly endless coefficients, significance tests, and summary statistics (Healy 2018; Schwabish 2021). Additionally, with the increasing ubiquity of network verbiage in society, they can even convey ideas in ways that require less "translation" work for audiences who sit outside of academic and research niches (Sayama et al. 2016; Harrington et al. 2013). Quite simply, network visualization is not only common, but can be incredibly useful (Birkett et al. 2021).[1]

Traditionally, this has taken the form of directly visualizing instances(s) of empirically **observed** social networks–whether snapshots of single data waves, or evolving network "movies" for dynamic relational data patterns (Ognyanova 2025).[2] That is, such visualizations are intended to characterize the data *per se*. While this practice predominates, it's worth asking: (1) why we visualize, and (2) what optimal set of principles govern strategies for accomplishing those aims, then (3) whether an alternative strategy serves those *purposes* and *needs* better than the existing approach?

In this paper, my answers to this set of questions suggests many practices of social network visualization are currently misaligned to their purposes–or more precisely, assumes a single visualization orientation can successfully address multiple (sets of) aims. As is often the case with split allegiances, I contend the predominant strategy leads us to occasionally miss the mark, especially when attempting to convey the *theoretical* takeaways of social networks research.

To address this misalignment, I propose that many (though notably, not all) purposes of social network visualizations would be better served by visualizing modeled networks rather than observed network data. After making this case following the motivations above, I illustrate this shift in three ways–a trivial example common to how many of us already teach social network modeling, describing a parallel strategy common in non-networks research, and an applied example from a recent research paper.

Before continuing, I should emphasize one thing–when I say "model" I intend this usage in a *theoretical* not a *statistical* sense. I avoid relegating this point to a footnote to highlight that the strategy I encourage does not only apply to statistically modeled results. Moreover, this general applicability with not only "extend" to applications outside of the recent proliferation of statistically modeled networks, but will also draw directly from other forms of SNA to demonstrate how elements of this strategy are already common practices in some circles (if labeled differently), and (partially) pave the way for how statistically oriented researchers can implement the approach.

Purposes of (network) visualization

So, what are the benefits of network data visualizations? Many of the benefits for network scholarship mirror those that motivate data visualization in science more broadly (Rawlings et al. 2023). A fundamental premise is that data visualization enhances **memory** (Schwabish 2021); this reflects what we know from a number of old adages–including that there's simply value in beauty Halpern (2015). Moreover, this memory benefit leverages the potential **efficiency** available in data visualization that simply cannot be achieved in other strategies for presenting evidence (Healy 2018). Edward Tufte has prioritized not only using this benefit, but attempting to find ways to optimize it, e.g., via efforts to maximize what he conceptualized as the "ink-to-information" ratio (Tufte 1983). These memory-oriented enhancements can arise both via improved recall and encoding processes–on the latter, the simple act of including visualizations can enhance memory via the complementarity of multimodal communication strategies, dovetailing with the strengths and weaknesses of text (Sweller 1988).

The data we use to represent social processes are almost always inherently high-dimensional, and visualizations provide a means for *simplifying* that complexity in structured, intepretable formats (Wickham et

[1]As with any data presentation strategy–visual or otherwise–there are also ample examples of how to do this poorly (e.g., what have come to be known as unintelligible network "hairballs"); I have a few published visuals of my own that I'd *substantially* rework if given the opportunity today.

[2]These typically occur *after* data pre-processing has taken place (e.g., cleaning, recoding, etc.), but no further data reduction steps (i.e., are not the product of data *summaries* or analytic interpretations).

al. 2023).[3] This leads visualizations to not only enhance communicative aims, but also to improve readers' ability to understand and interpret complex social phenomenon (Schwabish 2021). While statistics can summarize data in principled ways, data visualization rarely simply mimics these other forms and representations of data reduction. Instead, they are also "attention-drawing" efforts that highlight certain features of the data, while typically necessitating the deprioritization of others (Leifer 1992).

** How do network scholars *in particular* prioritize these combating aims? ** involve some claims about attention–drawing our attention to certain features of the data, and deprioritizing others .^{e.g., emphasizing the community structure of a graph in a }

Network scholars especially leverage these dimension reduction approaches by translating summaries of (theoretically) salient social space into the particular strategies for visualization forms (Rawlings et al. 2023). In practice, researchers assesses these aims differently depending on the stage of research in which they implement visualizations. In particular, optimal visualization strategies differ substantially between exploratory (Wickham et al. 2023), conceptual/theoretical (Brady et al. 2020), or explanatory aims (Healy and Moody 2014).

Given these potential differences, it is worth considering how readily existing uses implement different prioritization within visualizations. For example, if data fidelity, anomalous cases, and XXX are of particular interest in exploratory analyses, that would prioritize a set of *data* features as the key elements for visualization to highlight. In contrast, for conceptual development in formalizing the aims of a research question, visualization likely needs to optimize on elements of the *analytic strategy* - e.g., differentiating which components are explicitly examined, versus which are (unexamined) assumptions. Similarly, if being used as a strategy for presenting analytic results, the emphasis typically seeks to highlight *theoretical* implications of interpretation. Given the inherent trade-offs between data fidelity, theorized processes, and participant confidentiality (Moody et al. 2005), these prioritizations will frequently lead visualizations designed for one of these purposes serving poorly when leveraged for another. Perhaps more problematically, when researchers *assume* (or worse, *require*) that the aims and strategies are shared across these purposes, we can end up with visualizations that actually fail to accomplish any of these aims well (Pache and Santos 2010).

Principles for optimal (network) science visualization

This leads to a set of questions that should be addressed before any visualization is generated, namely - what are the desirable features of a visualization strategy? Here, I focus on three broad aims, with some differences within the precise implementations of these.

1. Given that researchers prioritize different sets of features to be highlighted in their visualizations– according to different phases of the research process, and these take on particular flavors in *network* visualizations. Exploratorily, network visualizations can serve to characterize a study population's composition and structural patterns, in ways that help to articulate things like: features that analyses seek to explain, or identify needed pre-processing steps (Wickham et al. 2023). Conceptually, network visualizations necessitate a concrete articulation of what analyses will and will not explicitly address, highlighting aspects of a theorized social and relational process to be evaluated (adams 2020). In social network research, the explanatory aims of these simplifications seek to characterize how readily the analytic results applied to the empirical case(s) observed (Kadushin 2012) when properly interpreted conform to the theoretical claims being modeled (Borgatti and Lopez-Kidwell 2011) – something subsequent visualizations can be optimized to clearly convey (Bender-deMoll and McFarland 2005). It is worth noting that while each of these research purposes leverage **simplifications** to achieve their aims, they do so in ways that induce non-overlapping prioritization in the types of visualizations each would produce.

This lack of overlap can be resolved in one of three ways. One strategy would be to develop visualization strategies that synthesize across these aims (e.g., plotting an observed network then overlaying features of

[3] Some contend that this is a *constraint* of data visualization imposed on data presentation strategies (by obfuscating its richness), rather than its benefit (Healy and Moody 2014).

the theoretical results – such as highlighting model results as features on top of those networks). A second alternative is to construct visualizations that prioritize one of these aims and assume it can carry benefits into the other uses. My critique is that we're often doing one of the 2 above, without having well-considered what the limitations of the selected approach are imposing. The third option is to generate (multiple) visualizations according to these separable purposes.

Optimizing *useful* simplifications that offer clarity of the theoretical claims being offered is therefore a fundamental principle governing (network) scientific data visualization.

- describe/explain differences need to be incorporated into the above.
- methodological over-simplification (assume the data are real, BUT, they're a characterization, and one with SUBSTANTIAL error.)

2. replicability

3. safety Some text from the previous iteration to incorporate here - One limitation that I contend especially needs to be more carefully considered among social network researchers. Namely, network visualizations often present unique ethical challenges to presenting research (cite Sage book). Especially in "closed population" settings (but not only there - see FB example), the potential for deductive disclosure is broad in any presentation of multiple variables (cite cs on this), and network data are uniquely primed to allow serial-revelations of participants' identities—i.e., the ability to identify one node in a graph (e.g., myself) often heightens the capacity of identifying others' identities in turn (e.g., my friends). To this end, our frequent practice of plotting empirical networks from data collection efforts may undermine principles of confidentiality and minimal risk in ways that should lead us to consider alternate visualization strategies that can simultaneously leverage these strengths while minimizing these potential risks (cite white paper). \textcolor{red}{Do I want to tell my story of hand coding the Add Health network from their 4-cycles paper to incorporate that model in my Glee paper?)

A synopsis of the traditional approach ?

1. obtain/clean data
2. plot & label
3. interpret

Plot Modeled, not Raw, Networks

From here still likely reads according to the old motivation for this piece - primarily ethical considerations, and may need some substantial revision yet to think broader about what's being solved. Especially should be recast as how/whether it accomplishes the list of aims elaborated above.

My fundamental proposition is simple - instead of plotting the "raw" empirical networks directly presenting data that has been collected and researchers in-turn analyze, we can produce visualizations of modeled network results that arise from those analyses. This is not a ground-breaking suggestion. In fact, it's a practice many statistical modelers of social network data already enact in practice. For example, "eye-balling" consistency of modeled results with observed networks–in tandem with formal statistical tests for goodness of fit–is even typically taught as a helpful practice in model development and fitting processes (Krivitsky et al. 2023). However, while we use these heuristic network visualizations to improve model fit (whether for capturing theoretical limitations, or for enhancing stability of estimates), when presenting networks in publications, authors still often resort to presenting the "real" network - often as a first-pass descriptive task to characterize the nature of the network(s) that the subsequent analyses are intended to explain. I've done this multiple times myself.

But if our aims in visualizing networks are to clearly, simply, systematically, and ethically convey scientifically useful network visualizations (recheck these parallel the way I end up ordering the set above.), there's no reason our visualizations need to be of the raw data. Modeled network representations can better serve our purposes across these aims.

Let me draw a parallel here. Logistic regression coefficients (just like exponential random graph models) are notoriously hard to make substantive sense of on their own (e.g., significance and magnitude of effects are frequently conflated in results interpretations by writers and readers alike). A common solution to this has been for researchers to: (1) gather, pre-process, then provide a descriptive summary of their data to given an overview sense of what the sample being analyzed actually represents, and how key focal features are distributed in the population; (2) conduct statistical analyses to investigate the researchers' theoretical aims, which are summarized with traditional combinations of betas, standard errors, log-likelihoods, etc. to represent the fit of those claims to empirical estimation, then (3) transform those modeled results into "predicted probabilities" in ways that help interpret what those modeled results "look like" in their implications (Fox and Andersen 2006). To "tell the story" that arises from the model results we don't return to the raw data, provide the individual-level combination of variables for some subset of the population and rely on those case's patterns to convey the implications of our research (in the rare cases that people do, they are often "synthetic" cases, not real ones, e.g., Ayala and Koch 2019). To do so would not only fail to clearly tell the aggregate story we've learned in the course of our theorization and analysis; it would also violate a number of human subjects protections researchers are obligated to follow. So why in the network world would we need to effectively present the "real" pattern of relationships from our data as a means to interpret and convey our findings in visual form?

Implementation

Instead, I suggest that we can follow a similar logic to develop presentation- and publication-worthy network visualizations from our modeled results in ways that accomplish the aims we have for visualizations–perhaps even more clearly than the directly observed versions–while maintaining protections against potential ethical gaps (e.g., deductive disclosure). The process I'm recommending follows the same logic as for regression-type models laid out above:

1. Present descriptives as *summary* statistics of the characteristics of one's network(s) being investigated (Neal et al. 2024).
2. Conduct whatever form of analyses are appropriate to the theoretical aims of the research question(s).
3. Compile appropriate analytic overviews of results (e.g., statistics) into summary form. Here I'm going to want to be especially clear that I'm using "model" as primarily a *theoretical* device/claim, not a particular form of (statistical) analysis. I.e., this doesn't have to be e.g., and ergm result.
4. Visualize "predicted" network(s) that are specified by the corresponding network analytic (model) results from which the researchers' conclusions are reached.

Worth noting that some corners of the field already effectively do this. Block images are a coarse-graining analytical result based strategy for presenting model characteristics (e.g., from a stochastic blockmodel), rather than the raw data from which they derive. Ego network visuals often use heuristic representations of their actual observations. Network motifs are an attempt to capture common features across a collection of networks, rather than presenting multiple parallel visualizations and attempting to extract those commonalities across the repeated panels.

Still to be filled in - Example

Work out the "aims" of visualization a bit more concretely, and elaborate how modeled viz may do this even better than raw data viz can.

. . .

Let's illustrate this with an example - I do think I want to use an ergm for this, especially given how frequently we already do this in the first place for that modeling framework as a means for interpretting/improving fit of model estimates.

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